

Search Frictions and Welfare in a Market for Prominence

Theory and Experiment

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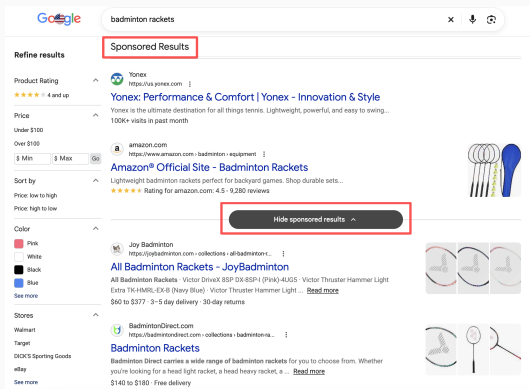
September 28, 2026

Practice Job Talk

Market for Prominence

Paid prominence: Sellers pay platforms to make products easier to discover and inspect.

- Example: Google search ads (2025: Google ad revenue worldwide, \$294.7 billion; U.S. search advertising revenue, \$114.2 billion)



Market for Prominence

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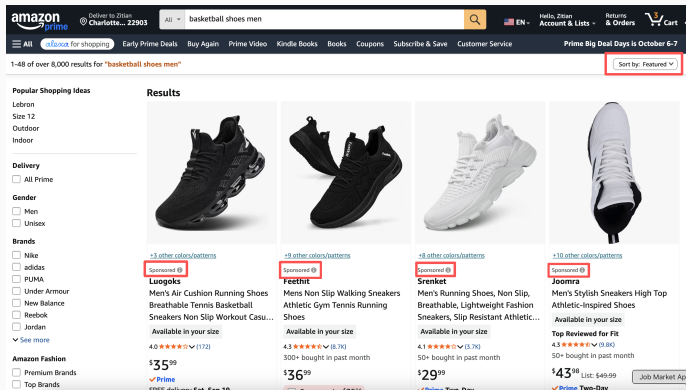
- Example: Expedia TravelAds

The screenshot shows the Expedia website interface for hotel bookings in Charlottesville, Virginia. The search parameters are set for a stay from Wednesday, September 16 to Friday, September 18, for 2 travelers in 1 room. The results page displays a list of hotels, including the English Inn of Charlottesville, DoubleTree by Hilton Hotel Charlottesville, and Super 8 by Wyndham Charlottesville. A red box highlights a 'TravelAd' for the English Inn of Charlottesville, which is a sponsored listing. The ad shows a rating of 4.5 stars and a price of \$222 nightly. The interface also includes a sidebar with filters for price, property type, and amenities, and a map view of the search area.

Market for Prominence

Paid prominence: Sellers pay platforms to make products easier to discover and inspect.

- Example: Amazon Sponsored Products



"70% of Amazon shoppers do not click past the first Search Results Page" (FTC complaint)

Search Friction and the Value of Prominence

Costly consumer search makes prominence valuable.

- Consumers may stop searching before inspecting every product.

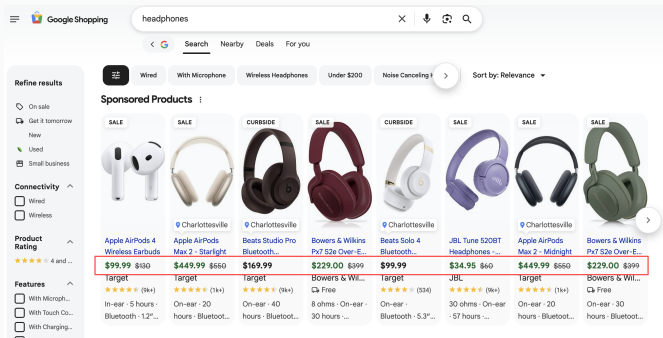
The value of prominence depends on how costly it is to inspect competing products.

- This relative advantage affects sellers' willingness to pay for prominence.

Visible Prices Guide Search

Price information is readily available, but finding a good fit remains costly.

- Examples: price advertising, price-comparison websites, price sorting



Implication: Sellers can use prices strategically to **direct** consumer search.

Research Question

How does **search friction** affect the **level and distribution of surplus** when sellers compete for prominence?

Methodology

- A model of competition for prominence, visible prices, and consumer search
- A second-price auction to implement a market for prominence
- A laboratory experiment varying the search cost: Low vs. High

What Does Prominence Mean?

The information content of search rankings is hard to identify in the field.

- Quality signals, personalized recommendations

In the lab, prominence is implemented as a structural **search-cost advantage**.

- The top position is free to inspect, the second is costly.

I vary search cost to study how the value of this advantage changes.

Related Literature

1. Search and pricing theory

- Diamond (1971), Weitzman (1979), Anderson & Renault (1999, 2018), Anderson et al. (2020), Obradovits & Plaickner (2023).

2. Paid prominence theory

- Armstrong et al. (2009), Armstrong & Zhou (2011), Athey & Ellison (2011), Xu et al. (2011), Anderson & Renault (2026).

3. Experiments on search and pricing

- Grether et al. (1988), Davis & Holt (1996), Abrams et al. (2000), Ursu (2018).

Contribution: Experimentally measure how search cost affects sellers' willingness to pay for prominence, with endogenous pricing and consumer search.

Roadmap

1. A Model of Search Prominence

- Game Setup
- Buyer's Search Rule
- Equilibrium Prices and Bids

2. Experimental Design

3. Experimental Results

- Welfare Decomposition
- Bids, Prices, and Search Behavior

4. Conclusion

Model Overview

Two sellers: *ex ante* identical, compete for a **top position** and then in the product market.

Position advantage: Inspecting the top seller is **free**, inspecting the second **costs** $s > 0$.

A buyer: unit demand, searches sequentially for a good match.

Horizontal differentiation: idiosyncratic taste for each seller's product.

– Match values are i.i.d. across positions:

$$v_k = \begin{cases} V, & \text{with prob. } q, \\ 0, & \text{with prob. } 1 - q, \end{cases} \quad k = 1 \text{ (top), } 2 \text{ (second).}$$

Three-Stage Game

Auction

Two sellers compete for a **top position** in a **second-price auction**.

Winner = position 1, pays losing bid $b^{(2)}$.

Loser = position 2, pays 0.

Pricing

After positions are assigned, sellers **simultaneously** choose prices (p_1, p_2) .
Sellers do not observe v_1 or v_2 .

Search

Buyer observes (p_1, p_2, v_1) **for free**.

May **pay a search cost** $s > 0$ to inspect the second seller and learn v_2 .
At any time, may purchase from any **inspected** seller (free recall), or exit (outside option of 0).

Seller profit = price charged $\cdot \mathbf{1}_{\text{sale}}$ – **losing bid** $\cdot \mathbf{1}_{\text{win auction}}$

Buyer utility = (match value – price paid) $\cdot \mathbf{1}_{\text{purchase}}$ – $s \cdot \mathbf{1}_{\text{search}}$

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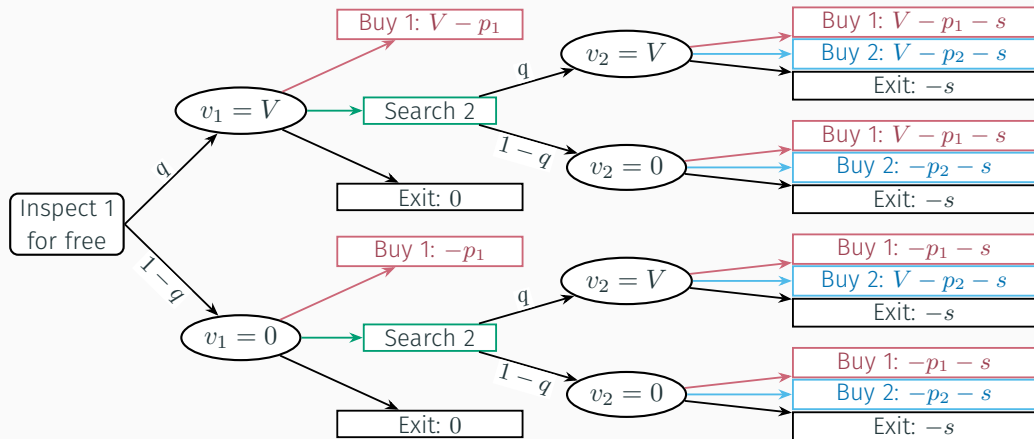
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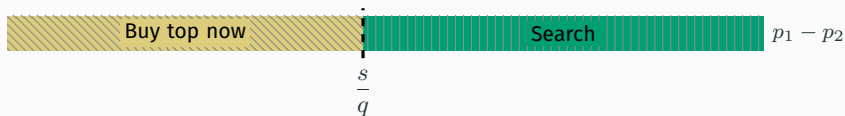
Buyer's Search Flow



Buyer's Search Rule

Observing (v_1, p_1, p_2) , the buyer compares the expected gain from search with its cost s .

- If $v_1 = V$: The buyer searches iff $q(p_1 - p_2) > s \iff p_1 - p_2 > \frac{s}{q}$

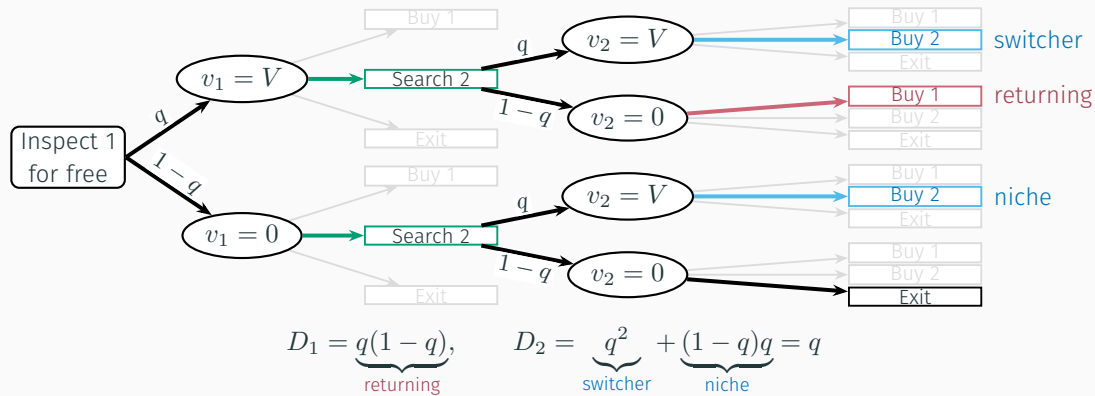


- If $v_1 = 0$: The buyer searches iff $q(V - p_2) \geq s \iff p_2 \leq V - \frac{s}{q}$



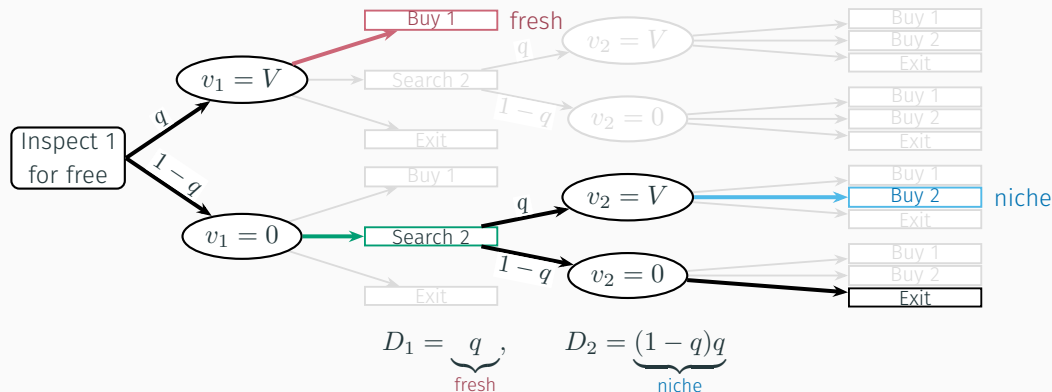
Search Activated for a Good-Top-Match Buyer

If $p_1 - p_2 > \frac{s}{q}$, a buyer with $v_1 = V$ **searches** (assuming $0 < p_1 \leq V$, $0 < p_2 \leq V - \frac{s}{q}$)



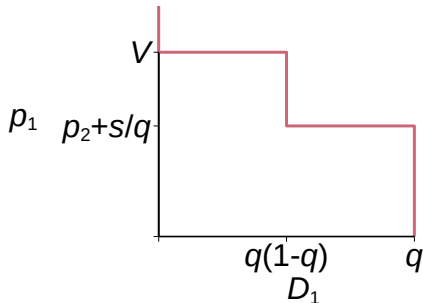
Search Deterred for a Good-Top-Match Buyer

If $p_1 - p_2 \leq \frac{s}{q}$, a buyer with $v_1 = V$ does **not** search (assuming $0 < p_1 \leq V$, $0 < p_2 \leq V - \frac{s}{q}$)

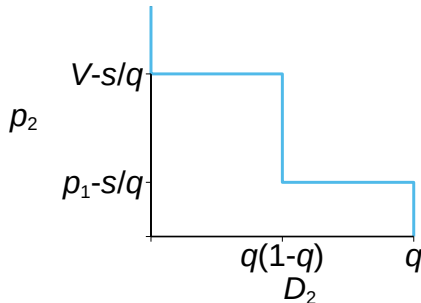


Demand Functions

Top seller



Second seller



No pure-strategy equilibrium due to opposing pricing incentives.

- **Exploitation** incentives due to **constant**-demand segments $\Rightarrow p \uparrow$
- **Undercutting** incentives due to **jumps** in demand $\Rightarrow p \downarrow$

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Mixed Strategy Pricing Equilibrium: Definition

For $p_1 \in (0, V]$ and $p_2 \in \left(0, V - \frac{s}{q}\right]$,

$$D_1(p_1, p_2) = \begin{cases} q(1-q), & p_1 - p_2 > \frac{s}{q} \\ q, & p_1 - p_2 \leq \frac{s}{q} \end{cases} \quad D_2(p_1, p_2) = \begin{cases} q, & p_1 - p_2 > \frac{s}{q} \\ q(1-q), & p_1 - p_2 \leq \frac{s}{q} \end{cases}$$

We seek distribution functions $F_1(p_1), F_2(p_2)$ with connected supports $[\underline{p}_1, \overline{p}_1]$ and $[\underline{p}_2, \overline{p}_2]$, s.t.

$$p_1^* \in \arg \max_{p_1 \in [0, +\infty)} \{p_1 \mathbb{E}_{F_2}[D_1(p_1, p_2)]\}, \quad \forall p_1^* \in [\underline{p}_1, \overline{p}_1]$$

$$p_2^* \in \arg \max_{p_2 \in [0, +\infty)} \{p_2 \mathbb{E}_{F_1}[D_2(p_1, p_2)]\}, \quad \forall p_2^* \in [\underline{p}_2, \overline{p}_2]$$

Mixed Strategy Pricing Equilibrium: Solution

For $0 < s < qV$, an equilibrium of the pricing subgame is:

$$F_1(p_1) = \frac{1}{q} \left(1 - \frac{(1-q) \left(V - \frac{s}{q} \right)}{p_1 - \frac{s}{q}} \right), \quad p_1 \in [\underline{p}_1, \overline{p}_1]$$

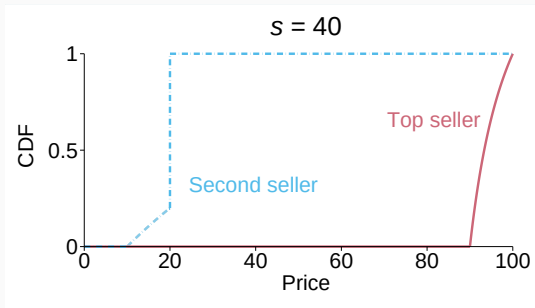
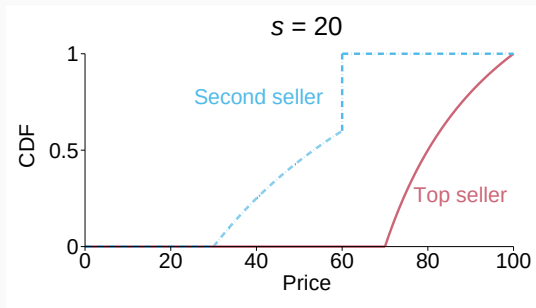
$$F_2(p_2) = \begin{cases} \frac{1}{q} \left(1 - \frac{(1-q)V + s}{p_2 + \frac{s}{q}} \right), & p_2 \in [\underline{p}_2, \overline{p}_2) \\ 1, & p_2 = \overline{p}_2 \end{cases}$$

$$\underline{p}_1 = (1-q)V + s, \quad \overline{p}_1 = V$$

$$\underline{p}_2 = (1-q) \left(V - \frac{s}{q} \right), \quad \overline{p}_2 = V - \frac{s}{q}$$

Seller 1 is atomless; seller 2 places a mass $\frac{s}{qV}$ at $\overline{p}_2$.

Mixed Strategy Pricing Equilibrium: Predictions



$$V = 100, q = 0.5$$

- **Price dispersion:** Both sellers mix over disjoint price intervals.
- **Price hierarchy:** p_1 *first-order stochastically dominates* p_2 .

Symmetric Bidding Equilibrium

Expected continuation values (revenues) for each position from the pricing subgame:

$$\Pi_1 = q[(1 - q)V + s], \quad \Pi_2 = q(1 - q) \left(V - \frac{s}{q} \right)$$

Equilibrium willingness to pay for prominence:

$$b^* = \Pi_1 - \Pi_2 = s$$

Profits are equalized across positions:

$$\underbrace{\Pi_1 - b^*}_{\pi_1} = \underbrace{\Pi_2}_{\pi_2}$$

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Experimental Design

Treatments (between-subject design):

- Low search cost: $s = 20$
- High search cost: $s = 40$
- Common parameters: $V = 100, q = 0.5$

Each market: 2 sellers + 1 buyer.

Each subject plays 30 rounds.

- Fixed roles; random rematching into markets anonymously in each round.
- Match values (v_1, v_2) are re-drawn each round.

Implementation

1. Feedback and payment
 - On-path market outcomes at the end of each round
 - \$10 show-up fee + earnings from one randomly selected round
2. Run 6 sessions at UVA Veconlab
 - **Low search cost:** 42 subjects from 3 sessions (12–15 per session)
 - **High search cost:** 45 subjects from 3 sessions (15 per session)
3. Number of observations (three-player markets):
 - **Low search cost:** 420 markets
 - **High search cost:** 450 markets

Sample Bidding Screen

Round 1 - Bidding

Bid for Top Position

You are a **Seller**. Please submit your bid for the Top Position on the buyer's screen. The seller with the higher bid becomes the **Top Seller**, and pays a Position Payment equal to the **Second Seller's bid**. The Second Seller pays 0 points.

(Reminder: The buyer observes everyone's prices for free. If the buyer chooses to visit the Second Seller to learn its product fit, they must pay a visit cost of 40 points. The Top Seller's product fit is always revealed for free.)

Your bid for the top position is:



34 points

Submit Bid

Sample Price-Setting Screen

Round 1 - Pricing

Bidding Results & Pricing

You are the Top Seller!

Your position payment for this round is **18 points**.

Please set the price for your product. Once posted, both sellers' prices will be **directly observed by the buyer** before they decide whether to visit the Second Seller.

Your price to be posted is:



65 points

Post Price

Sample Search Decision

Round 1 - Search

Market Overview

You are the **Buyer**. Sellers have posted their prices shown below.

You can see the product fit (worth 100 or 0) for the Top Seller for free.

Top Seller

65 points

Posted Price

Good Fit (100)

Product Fit

Second Seller

27 points

Posted Price

?

Product Fit

Decision

Do you want to pay **40 points** to visit the Second Seller and reveal their product fit?

If you don't visit, you can purchase from the Top Seller or quit. If you do visit, your 40 points are sunk, but you can purchase from either seller or quit later.

Visit Second Seller (Pay 40)

Buy from Top Seller Now

Quit (Buy Nothing)

Back to game

Back to flow

Sample Purchase After Search

Round 1 - Purchase

Second Seller Revealed

You paid the visit cost. The product fit for the Second Seller has been revealed.

Top Seller

65 points

Posted Price

Good Fit (100)

Product Fit

Purchase

Second Seller

27 points

Posted Price

Good Fit (100)

Product Fit

Purchase

Quit (Buy Nothing)

Sample Round-End Feedback

Round 1 - Round Results

Round Summary

Market Outcomes

Position	Price Posted	Product Fit
Top Seller	65	Good Fit (100)
Second Seller	27	Good Fit (100)

Buyer's Decision

- **Visited Second Seller?** Yes (Paid 40)
- **Purchased From:** Second Seller

Payoff Points Earned This Round

$100 - 27 - 40 = +33 \text{ points}$

starting (200) + earned (33 points) = 233 points

Reminder: This formal round could potentially be selected to determine your final earnings.

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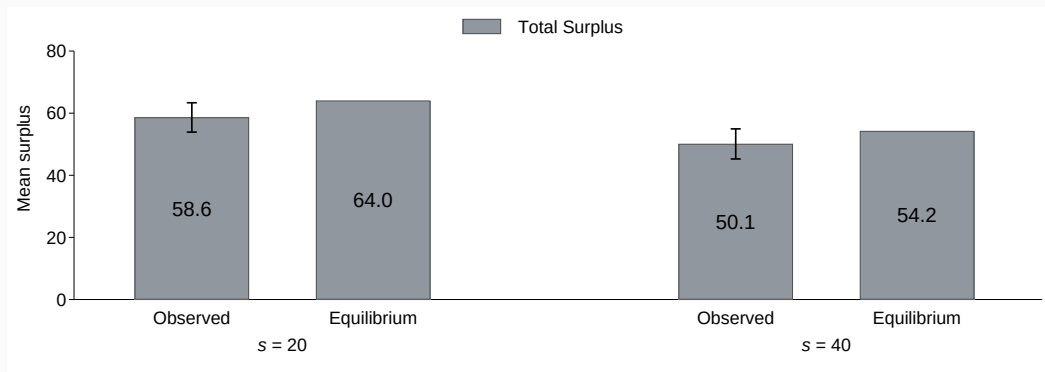
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- **Welfare Decomposition**
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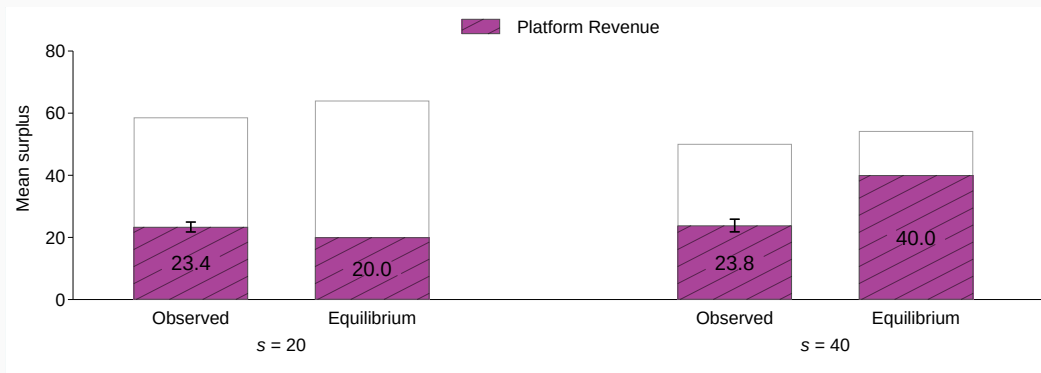
Total Surplus



Observed TS **decreases** by 8.5 from $s = 20$ to $s = 40$ ($p = 0.013$).

Observed TS falls **below** equilibrium at both search costs (paired $p < 0.001$, $p = 0.004$).

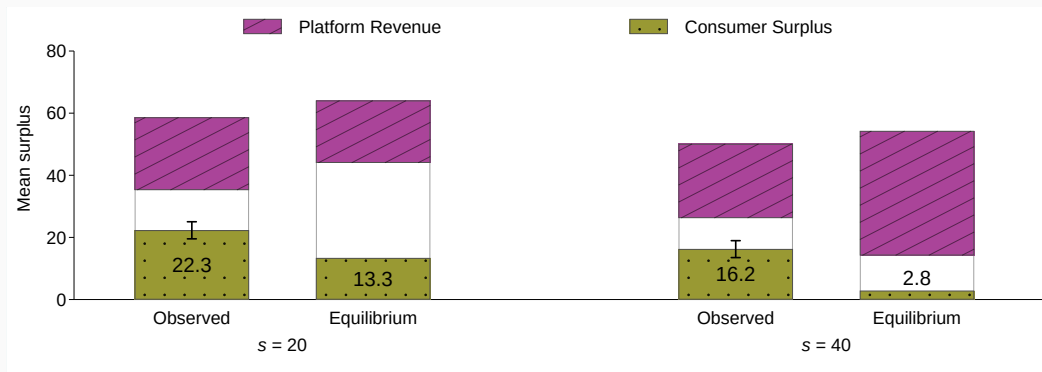
Platform Revenue



Observed PR **does not differ** significantly between $s = 20$ and $s = 40$.

Observed PR is **above** equilibrium at $s = 20$ and **below** at $s = 40$ (both $p < 0.001$).

Consumer Surplus

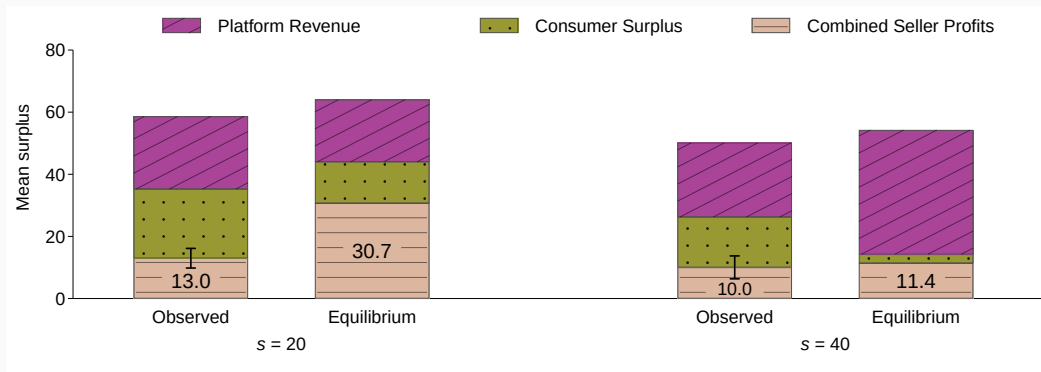


Observed CS **exceeds** equilibrium at both search costs (both paired $p < 0.001$).

Observed CS **decreases** by 6.1 from $s = 20$ to $s = 40$ ($p = 0.002$).

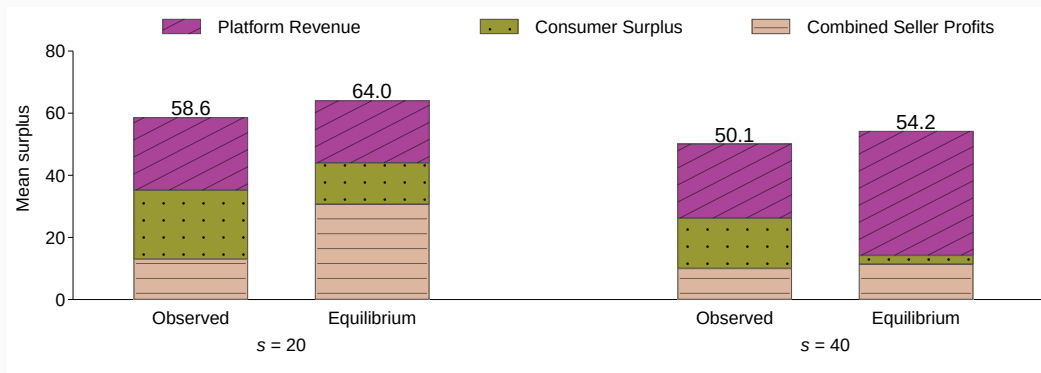
Observed CS **falls** by 27.2%, lower than the predicted 78.7% ($p < 0.001$).

Combined Seller Profits



Profits are **below** equilibrium at $s = 20$ (paired $p < 0.001$) and close to equilibrium at $s = 40$. Profits account for only around 20% of TS at both search costs.

Welfare Decomposition: Takeaways



1. Platform revenue changes little with search cost, despite the predicted increase.
2. Consumer surplus exceeds predictions at both costs and falls proportionally less.
3. Sellers jointly retain around 20% of total surplus after paying for prominence.

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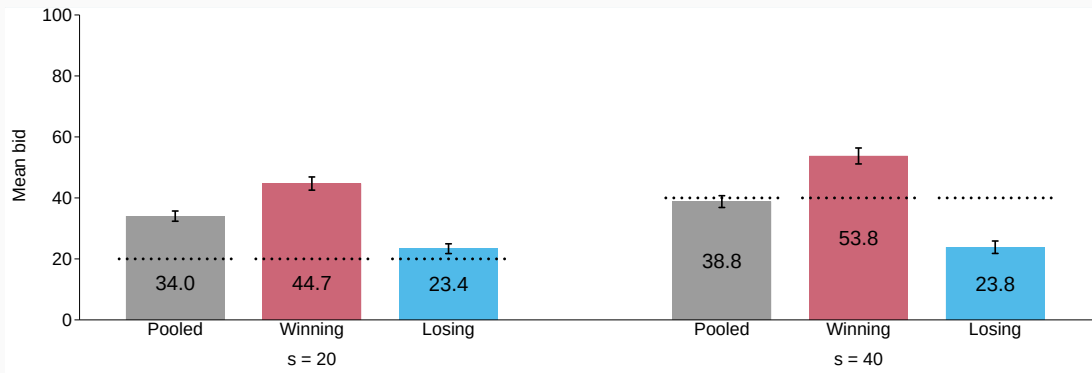
3. Experimental Results

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1. Platform revenue changes little with search cost, despite the predicted increase.

Observed Bids



Losing bids **do not differ** significantly between $s = 20$ and $s = 40$.

Pooled bids **increase** by 4.8 ($p < 0.001$), far less than the predicted increase of 20 ($p < 0.001$).

Explaining Welfare: Platform Revenue

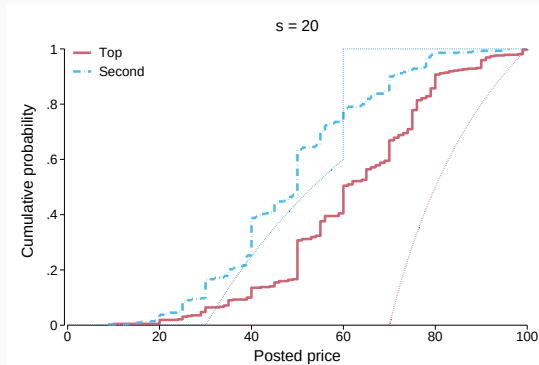
1. Platform revenue changes little with search cost, despite the predicted increase.

The losing bid (=platform revenue) is insensitive to the search cost.

Explaining Welfare: Consumers and Sellers

2. Consumer surplus exceeds predictions at both costs and falls proportionally less.
3. Sellers jointly retain around 20% of total surplus after paying for prominence.

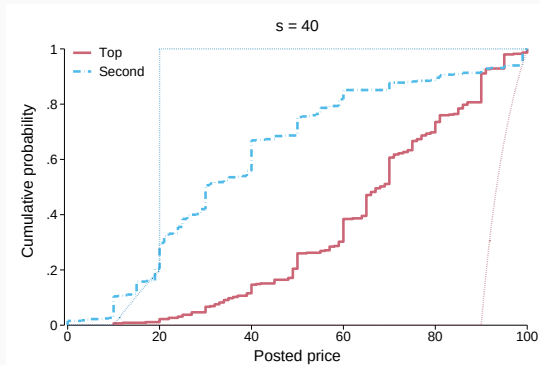
Price Distributions by Position



p_1, p_2 are more dispersed than predicted.

$p_1 > p_2$ (one-sided KS: $p < 0.001$).

Mean p_1 is below equilibrium ($p < 0.001$).



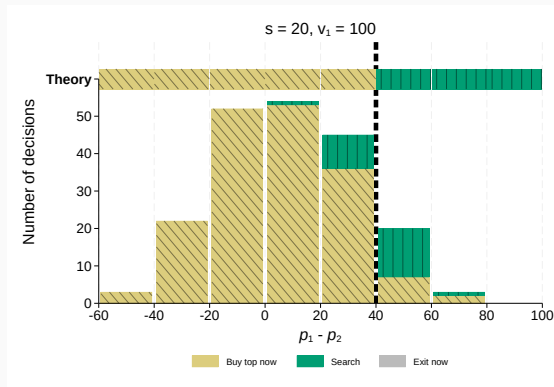
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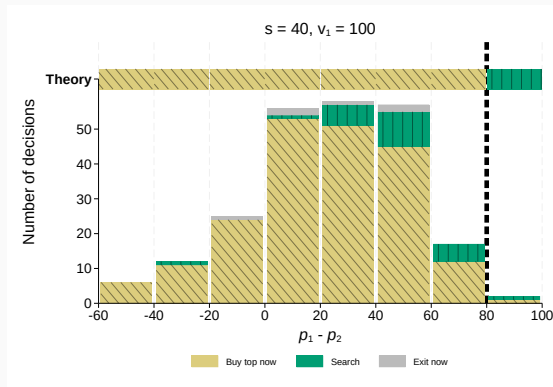
Mean p_1 is below equilibrium ($p < 0.001$).

Mean p_2 is above equilibrium ($p < 0.001$).

Search Decisions Given a Good Top Match



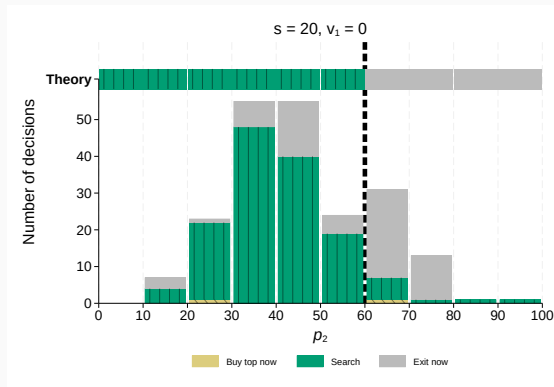
Compliance = 90.5%.



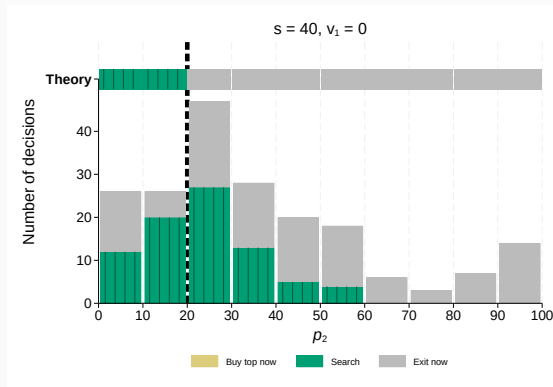
Compliance = 89.7%.

Most search decisions follow the theoretical cutoff at both search costs.

Search Decisions Given a Bad Top Match



Compliance = 80.5%.



Compliance = 64.6%.

Compliance falls from $s = 20$ to $s = 40$ ($p = 0.024$),
with both under- and over-search errors around and away from the cutoffs.

Explaining Welfare: Consumers and Sellers

2. Consumer surplus exceeds predictions at both costs and falls proportionally less.

Lower top prices make purchases from the top seller cheaper for consumers.

3. Sellers jointly retain around 20% of total surplus after paying for prominence.

Lower top prices reduce revenue per sale, while high second-seller prices discourage search after a bad top match and lead to lost sales.

Sellers earn less revenue than predicted and pay a substantial share to platform.

Conclusion

1. A model of competition for prominence, pricing, and consumer search.
2. An experiment varying search cost to study the level and distribution of surplus.
3. Bidding, pricing, and search behavior help explain the welfare findings:
 - Platform revenue changes little with search cost, reflecting stable losing bids.
 - Consumer surplus exceeds predictions, supported by lower top prices.
 - Sellers retain around 20% of total surplus: Lower top prices and lost sales reduce revenue, and position payments further reduce profits.

1. The platform's incentives to manipulate search cost.
2. Should prominence be allocated through market mechanisms?
 - Random allocation; other auction formats, e.g. first-price
 - $N > 2$ sellers, with multiple tiers of prominence ($0 < s_1 < s_2 < s_3 < \dots$)
3. Regulating how sellers direct search: Which product characteristics should sellers be allowed to display to consumers?

Thank you!

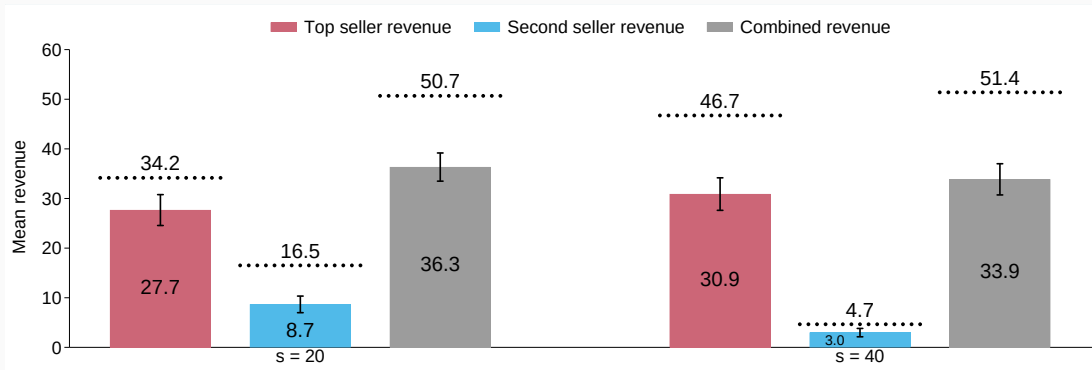
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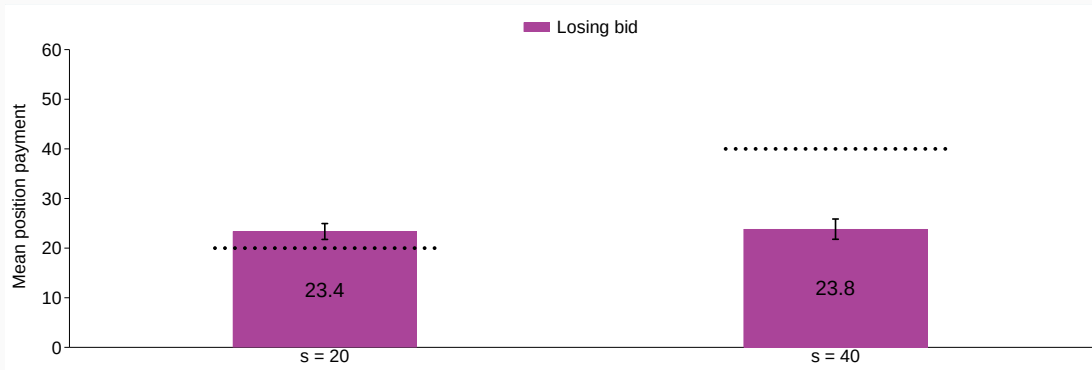
economics.virginia.edu/veconlab

Seller Revenue: By Position and Combined



Seller revenue, by position and combined, falls **below** equilibrium at both search costs (all paired $p < 0.001$).

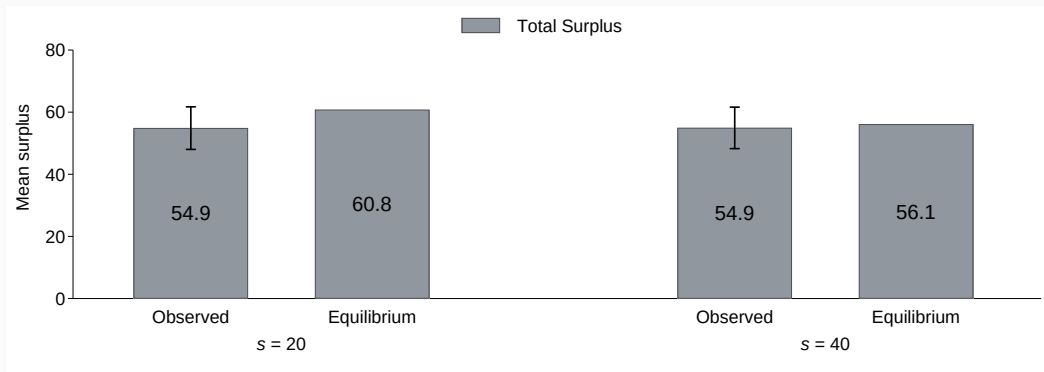
Position Payments



At $s = 20$, position payments **exceed** equilibrium by 3.4 ($p < 0.001$).

At $s = 40$, position payments fall **below** equilibrium by 16.2 ($p < 0.001$).

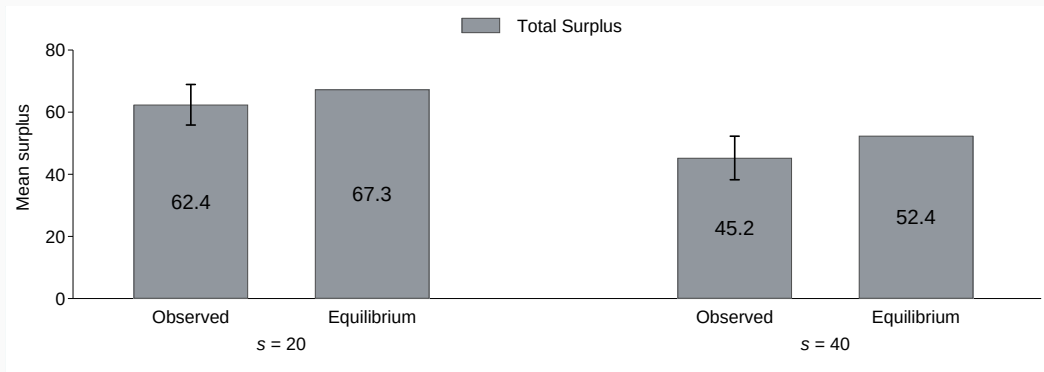
Total Surplus: Rounds 1–15



Observed TS **does not differ** significantly between $s = 20$ and $s = 40$.

Observed TS is **below** equilibrium at $s = 20$ (paired $p = 0.001$), but not significantly at $s = 40$.

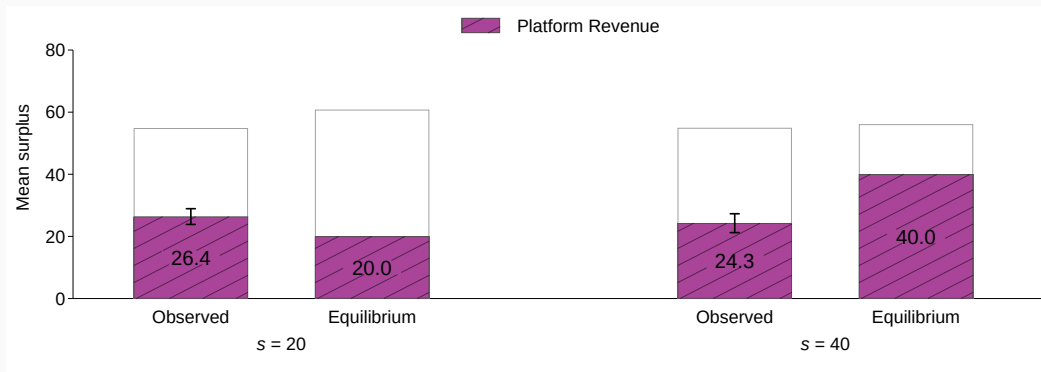
Total Surplus: Rounds 16–30



Observed TS **decreases** by 17.1 from $s = 20$ to $s = 40$ ($p < 0.001$).

Observed TS falls **below** equilibrium at both search costs (paired $p = 0.008$, $p = 0.002$).

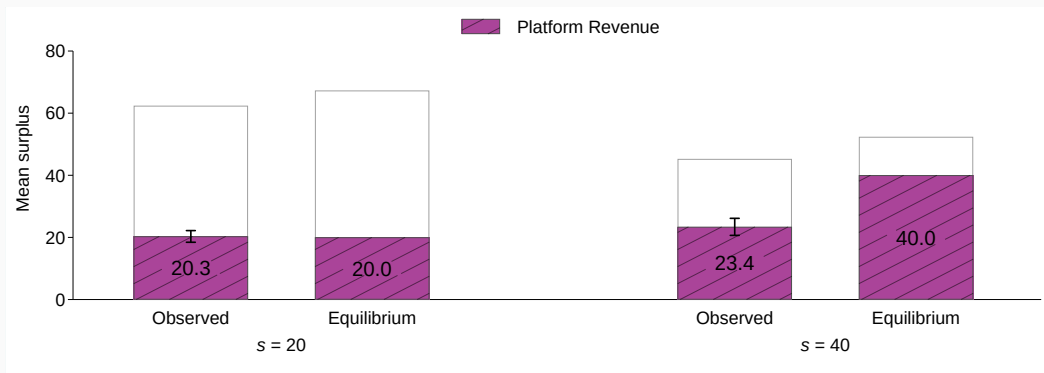
Platform Revenue: Rounds 1–15



Observed PR **does not differ** significantly between $s = 20$ and $s = 40$.

Observed PR is **above** equilibrium at $s = 20$ and **below** at $s = 40$ (both $p < 0.001$).

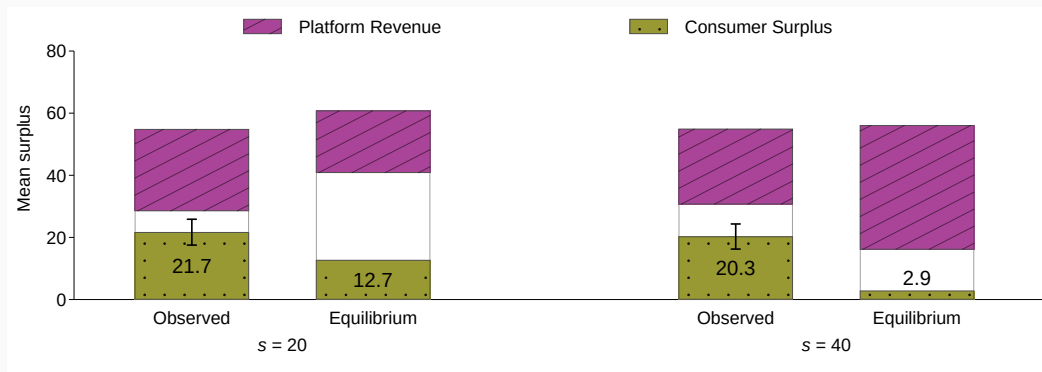
Platform Revenue: Rounds 16–30



Observed PR **does not differ** significantly between $s = 20$ and $s = 40$.

Observed PR is close to equilibrium at $s = 20$ and **below** at $s = 40$ ($p < 0.001$).

Consumer Surplus: Rounds 1–15

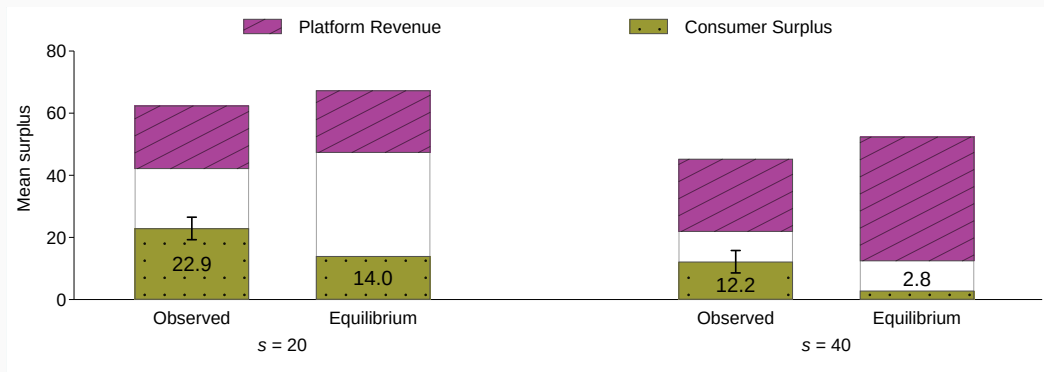


Observed CS **exceeds** equilibrium at both search costs (both paired $p < 0.001$).

Observed CS **decreases** by 1.4 from $s = 20$ to $s = 40$ (not significant).

Observed CS **falls** by 6.5%, lower than the predicted 77.6% ($p < 0.001$).

Consumer Surplus: Rounds 16–30

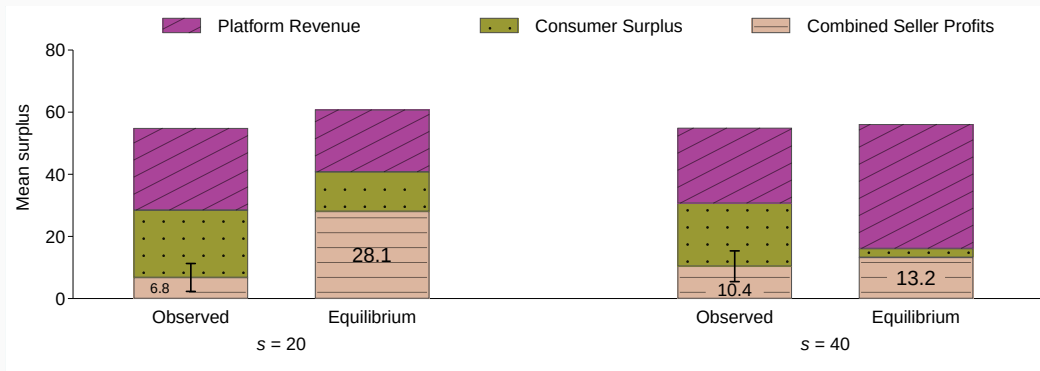


Observed CS **exceeds** equilibrium at both search costs (both paired $p < 0.001$).

Observed CS **decreases** by 10.7 from $s = 20$ to $s = 40$ ($p < 0.001$).

Observed CS **falls** by 46.8%, lower than the predicted 79.8% ($p = 0.013$).

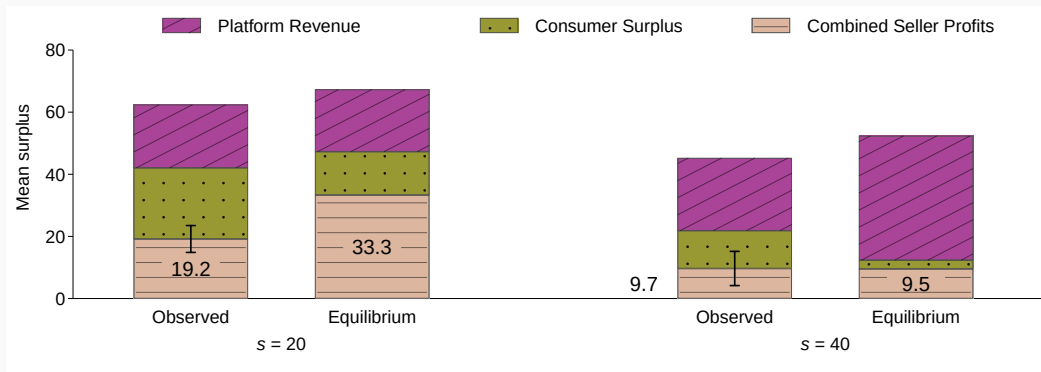
Combined Seller Profits: Rounds 1–15



Profits are **below** equilibrium at $s = 20$ (paired $p < 0.001$).

Profits account for around 12% of TS at $s = 20$ and 19% at $s = 40$.

Combined Seller Profits: Rounds 16–30

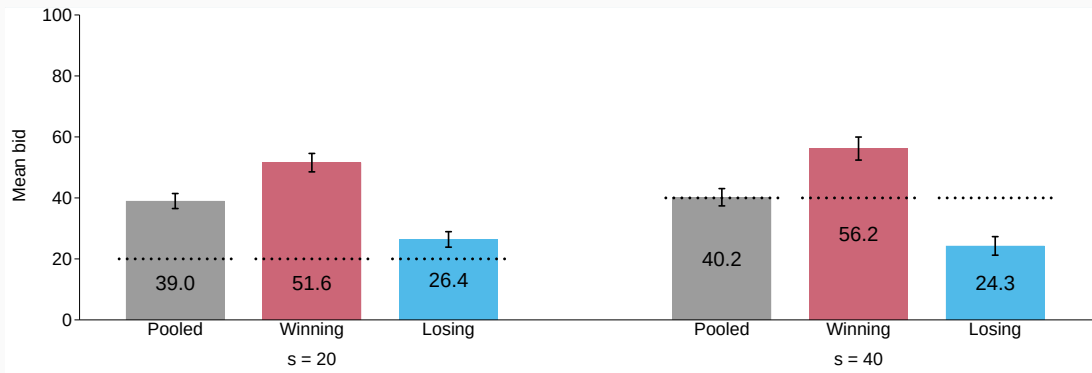


Profits are **below** equilibrium at $s = 20$ (paired $p < 0.001$).

Profits account for around 31% of TS at $s = 20$ and 21% at $s = 40$.

Profits **decrease** by 9.5 from $s = 20$ to $s = 40$ ($p = 0.008$).

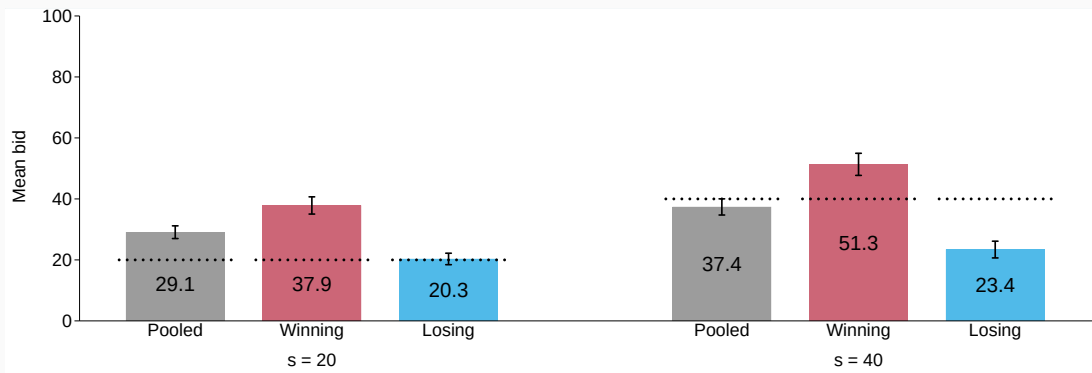
Observed Bids: Rounds 1–15



Losing bids **do not differ** significantly between $s = 20$ and $s = 40$.

Pooled bids **increase** by 1.2 (not significant), far less than the predicted 20 ($p < 0.001$).

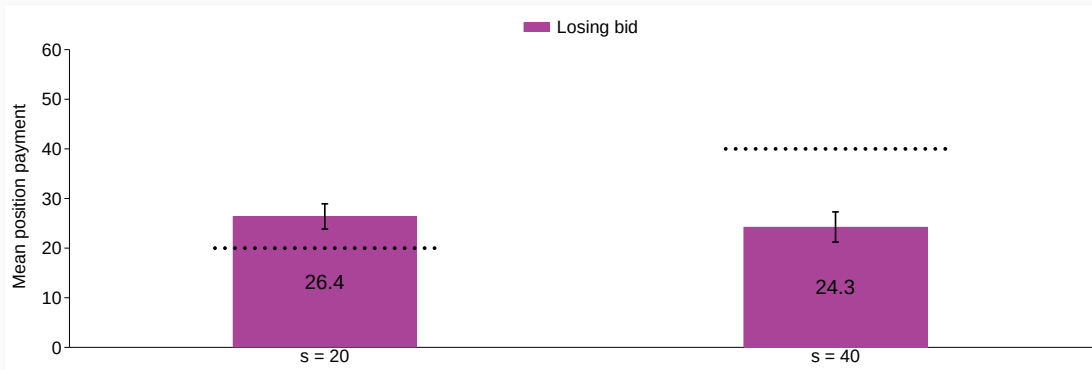
Observed Bids: Rounds 16–30



Losing bids **do not differ** significantly between $s = 20$ and $s = 40$.

Pooled bids **increase** by 8.3 ($p < 0.001$), far less than the predicted increase of 20 ($p < 0.001$).

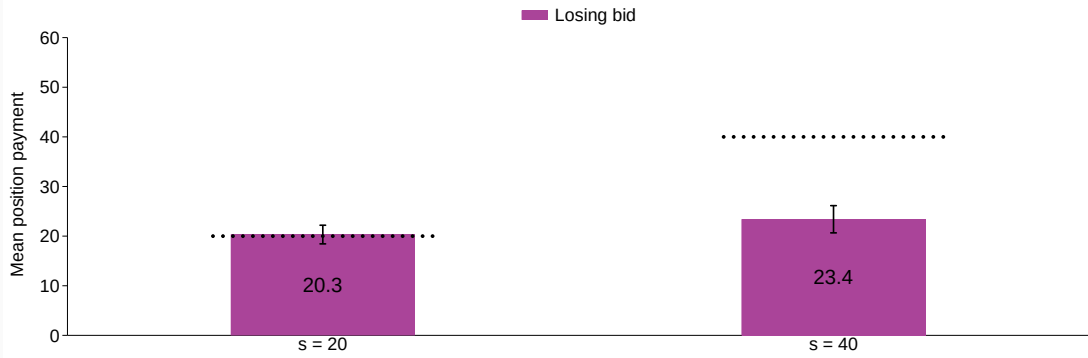
Position Payments: Rounds 1–15



At $s = 20$, position payments **exceed** equilibrium by 6.4 ($p < 0.001$).

At $s = 40$, position payments fall **below** equilibrium by 15.7 ($p < 0.001$).

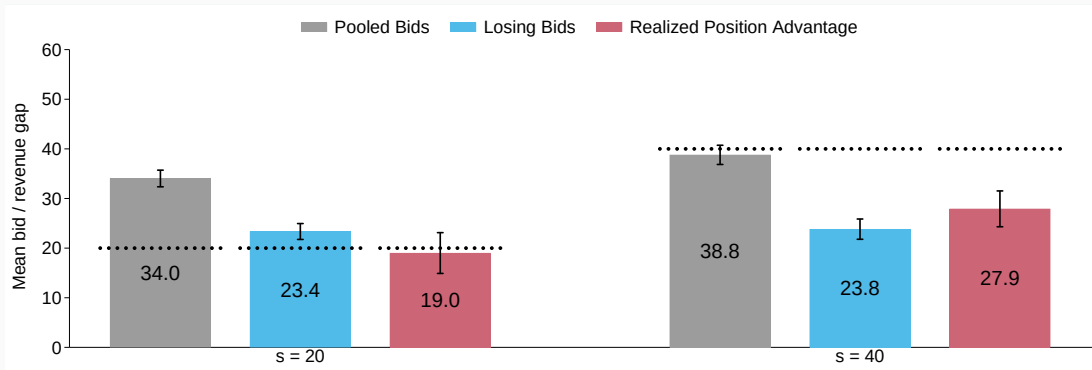
Position Payments: Rounds 16–30



At $s = 20$, position payments **do not differ** significantly from equilibrium.

At $s = 40$, position payments fall **below** equilibrium by 16.6 ($p < 0.001$).

Bids, Payments, and Position Advantage

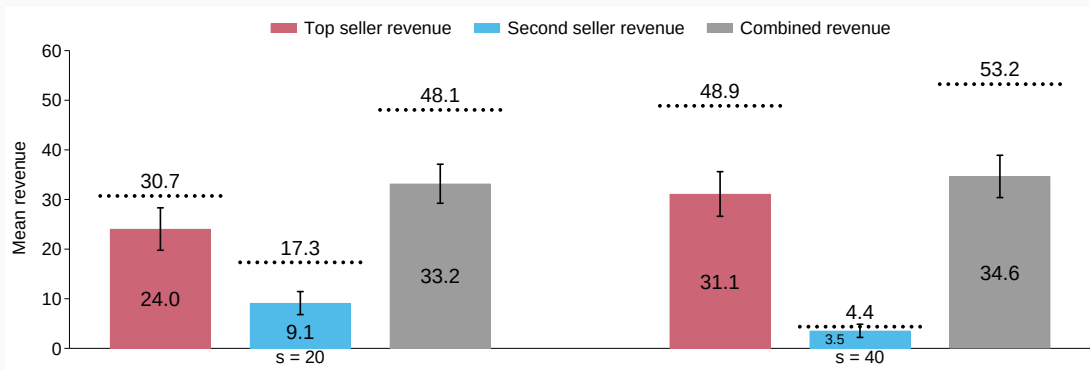


Realized position advantage is the observed revenue difference across positions $= \Pi_1 - \Pi_2$.

Pooled bids **exceed** this advantage at both search costs (both paired $p < 0.001$).

Payments **exceed** it at $s = 20$ (paired $p = 0.047$), but fall **below** it at $s = 40$ (paired $p = 0.049$).

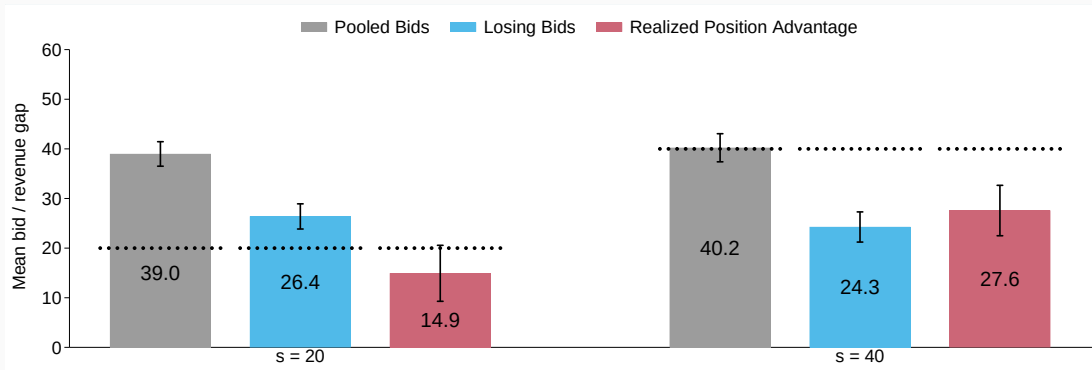
Seller Revenue: By Position and Combined, Rounds 1–15



Top seller revenue and combined revenue fall **below** equilibrium at both search costs (all paired $p < 0.001$).

Second seller revenue falls **below** equilibrium at $s = 20$ (paired $p < 0.001$).

Bids, Payments, and Position Advantage: Rounds 1–15

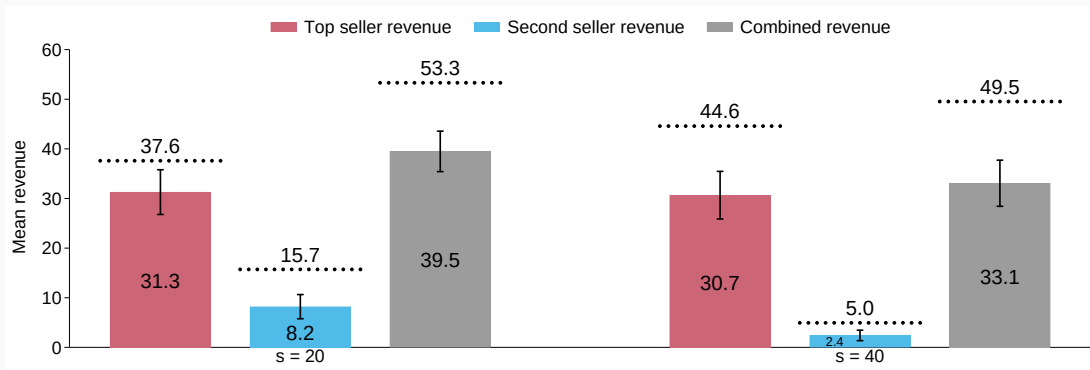


Realized position advantage is the observed revenue difference across positions $= \Pi_1 - \Pi_2$.

Pooled bids **exceed** this advantage at both search costs (both paired $p < 0.001$).

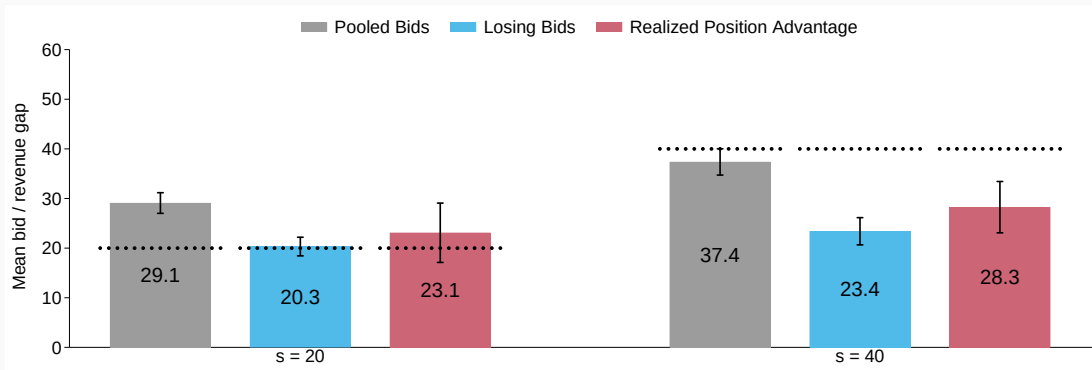
Payments **exceed** this advantage significantly only at $s = 20$ (paired $p < 0.001$).

Seller Revenue: By Position and Combined, Rounds 16–30



Seller revenue, by position and combined, falls **below** equilibrium at both search costs (all paired $p < 0.001$).

Bids, Payments, and Position Advantage: Rounds 16–30

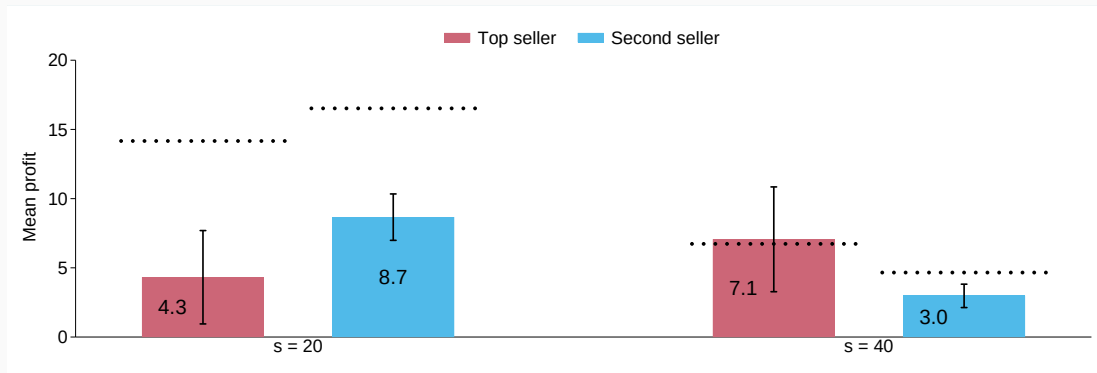


Realized position advantage is the observed revenue difference across positions $= \Pi_1 - \Pi_2$.

Pooled bids **exceed** this advantage significantly only at $s = 40$ (paired $p = 0.002$).

Payments **do not differ** significantly from it at either search cost.

Seller Profit by Position



Top-seller profit **does not differ** significantly between $s = 20$ and $s = 40$.

Second-seller profit **falls** from $s = 20$ to $s = 40$ ($p < 0.001$).

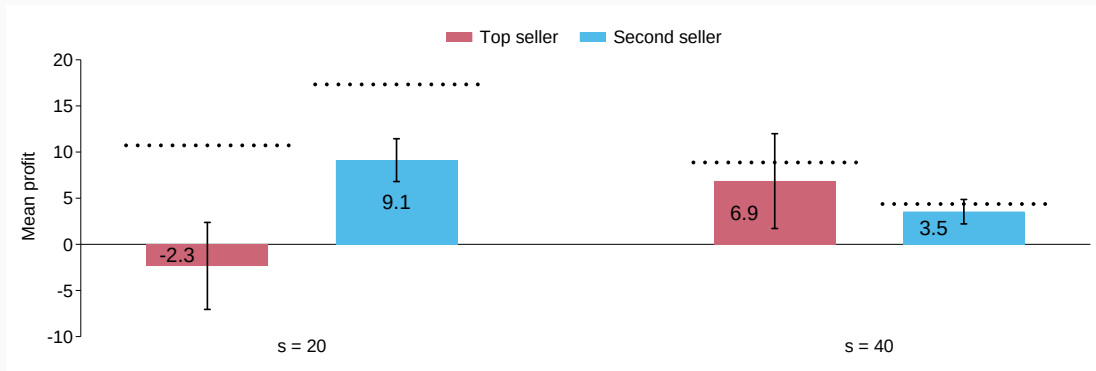
Combined profits **do not differ** significantly between $s = 20$ and $s = 40$.

[Rounds 1–15](#)

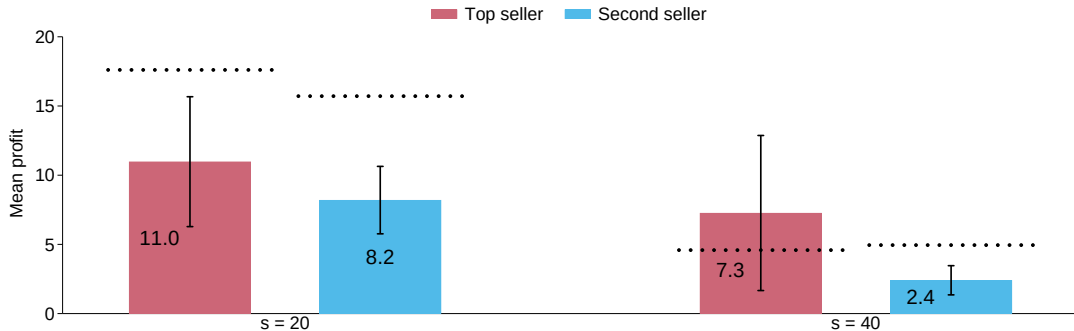
[Rounds 16–30](#)

[Back to Combined Seller Profits](#)

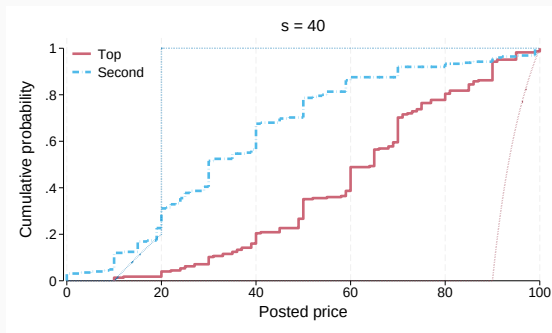
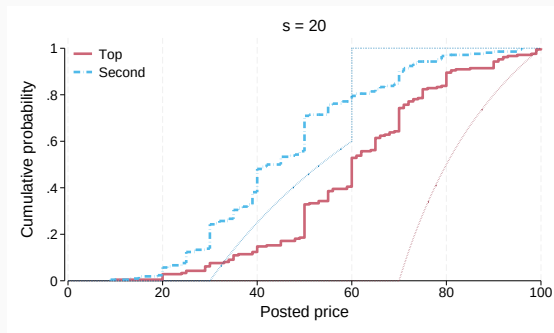
Seller Profit by Position: Rounds 1–15



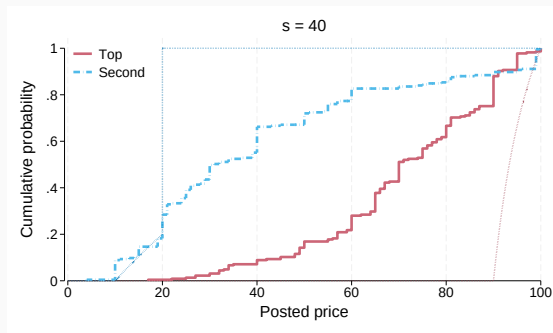
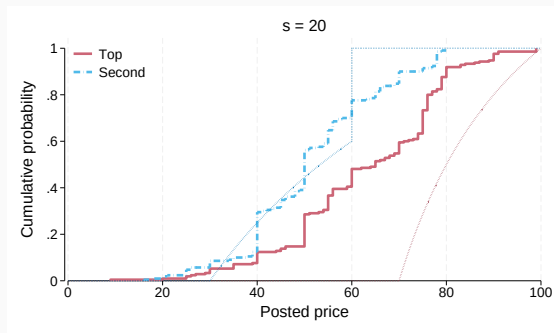
Seller Profit by Position: Rounds 16–30



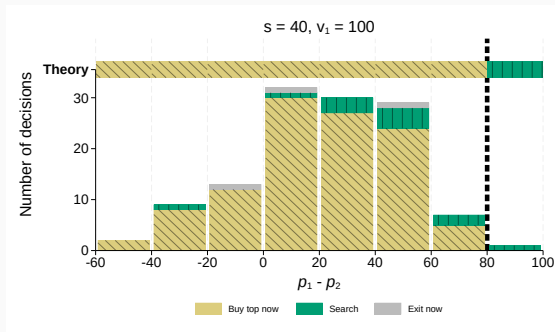
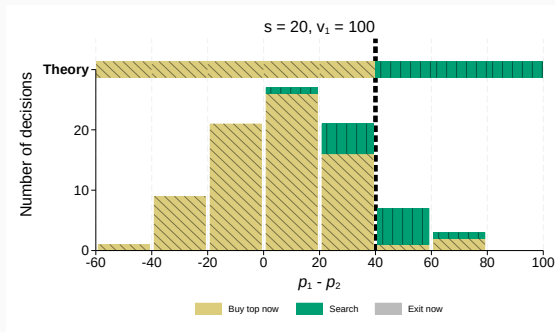
Price Distributions by Position: Rounds 1–15



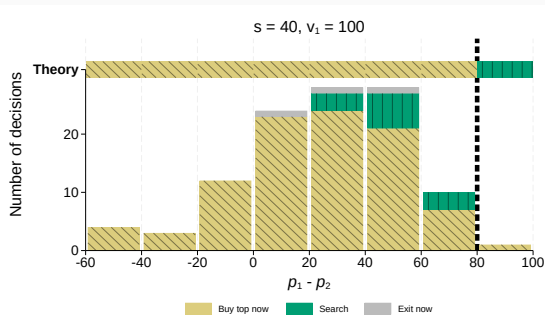
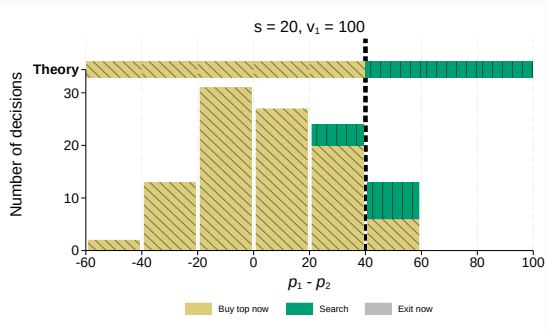
Price Distributions by Position: Rounds 16–30



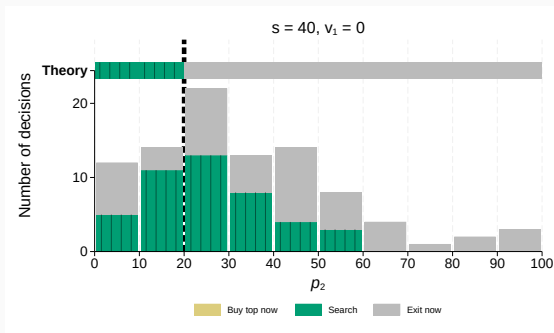
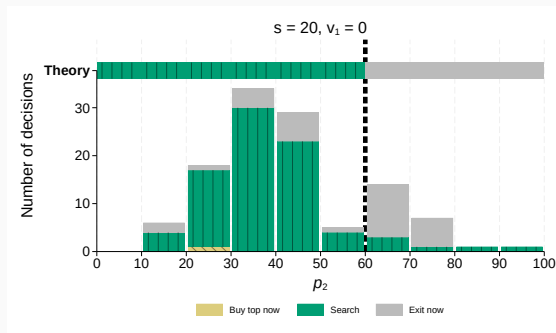
Search Decisions Given a Good Top Match: Rounds 1–15



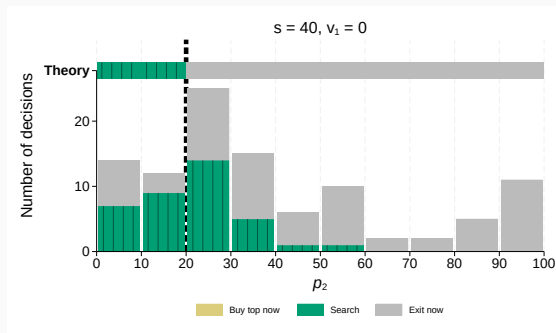
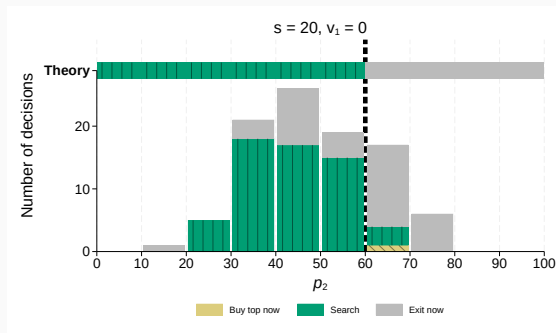
Search Decisions Given a Good Top Match: Rounds 16–30



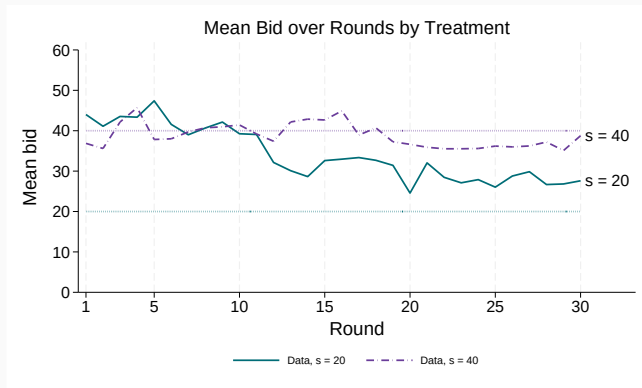
Search Decisions Given a Bad Top Match: Rounds 1–15



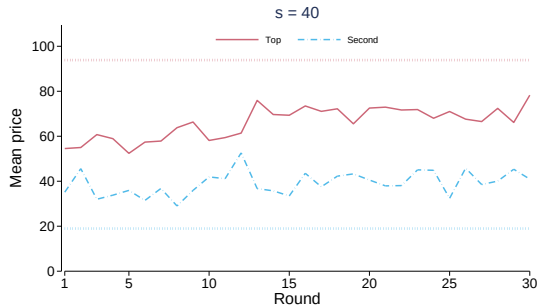
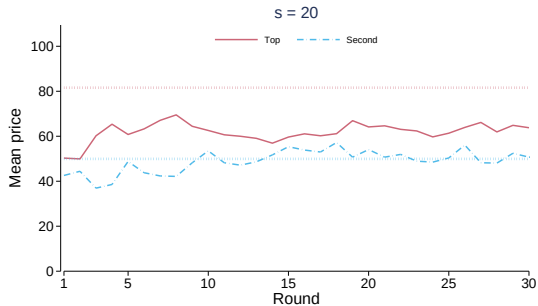
Search Decisions Given a Bad Top Match: Rounds 16–30



Bids over Rounds



Prices over Rounds



Summary of Predictions

$V = 100, q = 0.5$	$s = 20$	$s = 40$
Top price support	[70, 100]	[90, 100]
Second price support	[30, 60]	[10, 20]
Bids $b^* = \Pi_1 - \Pi_2 = s$	20	40
Under $v_1 = V$: Search cutoff in $p_1 - p_2$	40	80
Under $v_1 = 0$: Search cutoff in p_2	60	20
<i>Ex-ante outcomes</i>		
Top revenue Π_1	35	45
Second revenue Π_2	15	5
Profit $\pi_1 = \pi_2$	15	5
Consumer surplus	12.34	3.39
Platform revenue	20	40